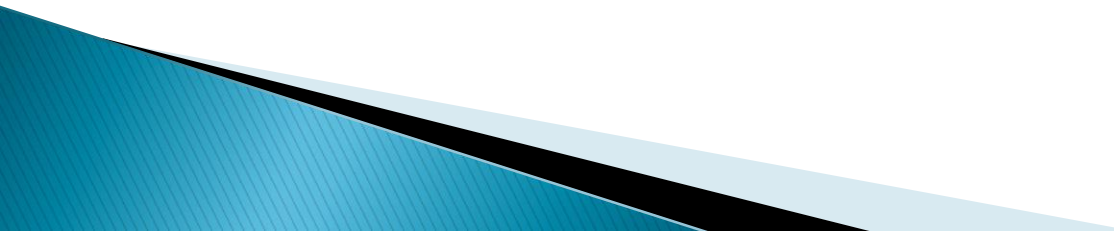


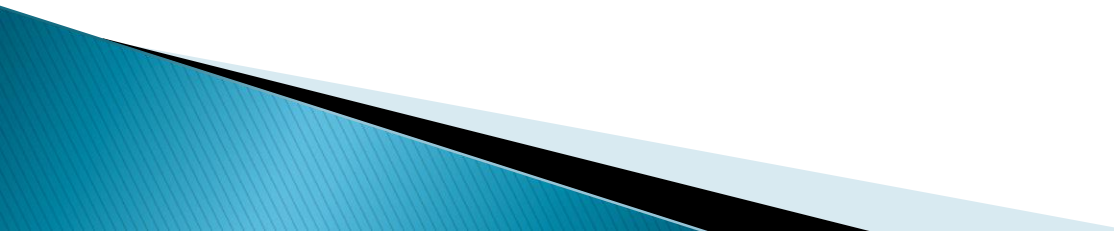
Introduction to Algorithms



Algorithms:

- ▶ Step by step method of solving problem.
 - ▶ The word – “Algorithm” derives from the name of the 9th Century Persian Mathematician – “Al – khowarizimi”.
 - ▶ Step by step means one step is followed by the other step. i.e. Previous step is pre – requisite of the next step.
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Properties – An Algorithm must have:

- ▶ Input
 - ▶ Output
 - ▶ Precision
 - ▶ Determinism
 - ▶ Finiteness
 - ▶ Correctness
 - ▶ Generality
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Input & Output

- ▶ An algorithm must have something as Input.
- ▶ It should also give some as Output

Precision

- ▶ Steps, written in the algorithm, should be precisely stated.

Determinism

- ▶ The intermediate results of each step, of execution, are unique and are determined only by inputs and results of the preceding steps.

Finiteness

- ▶ Algorithm should terminate.
- ▶ It means the algorithm should stop after finitely many instructions have been executed.

Correctness

- ▶ Output produced by the Algorithm should be correct.

Generality

- ▶ Algorithm should be generalized in nature.
- ▶ It should apply to a set of Input.

Example

- ▶ To find the Largest among three Numbers (a,b,c)

1. $x = a$
2. If $b > x$ then $x = b$
3. If $c > x$ then $x = c$
4. Print x

The above procedure (4 steps) is perfect example algorithm. It has all properties.

Big Oh

- ▶ Let f and g be non negative function on the +ve integers.
- ▶ We write $f(n) = O(g(n))$
 - i.e. $f(n)$ is of order, at most $g(n)$
 - or
 - $f(n)$ is *big oh* of $g(n)$
 - If there exist constants C_1 and N_1 such that $f(n) \leq C_1 g(n)$ for all $n \geq N_1$
 - i.e. f is bounded above by g
 - or g is an asymptotic upper bound for f

Omega (Ω)

- ▶ Let f and g be non negative function on the +ve integers.
- ▶ We write $f(n) = \Omega(g(n))$
 - i.e. $f(n)$ is of order, at least $g(n)$
 - or
 - $f(n)$ is *omega* of $g(n)$
 - If there exist constants C_2 and N_2 such that $f(n) \geq C_2 g(n)$ for all $n \geq N_2$
 - i.e. f is bounded below by g
 - or g is an asymptotic lower bound for f